Jan. 14, 2022

# CS 886 Deep Learning for Biotechnology

Ming Li

# CONTENT


# Attention and Transformers

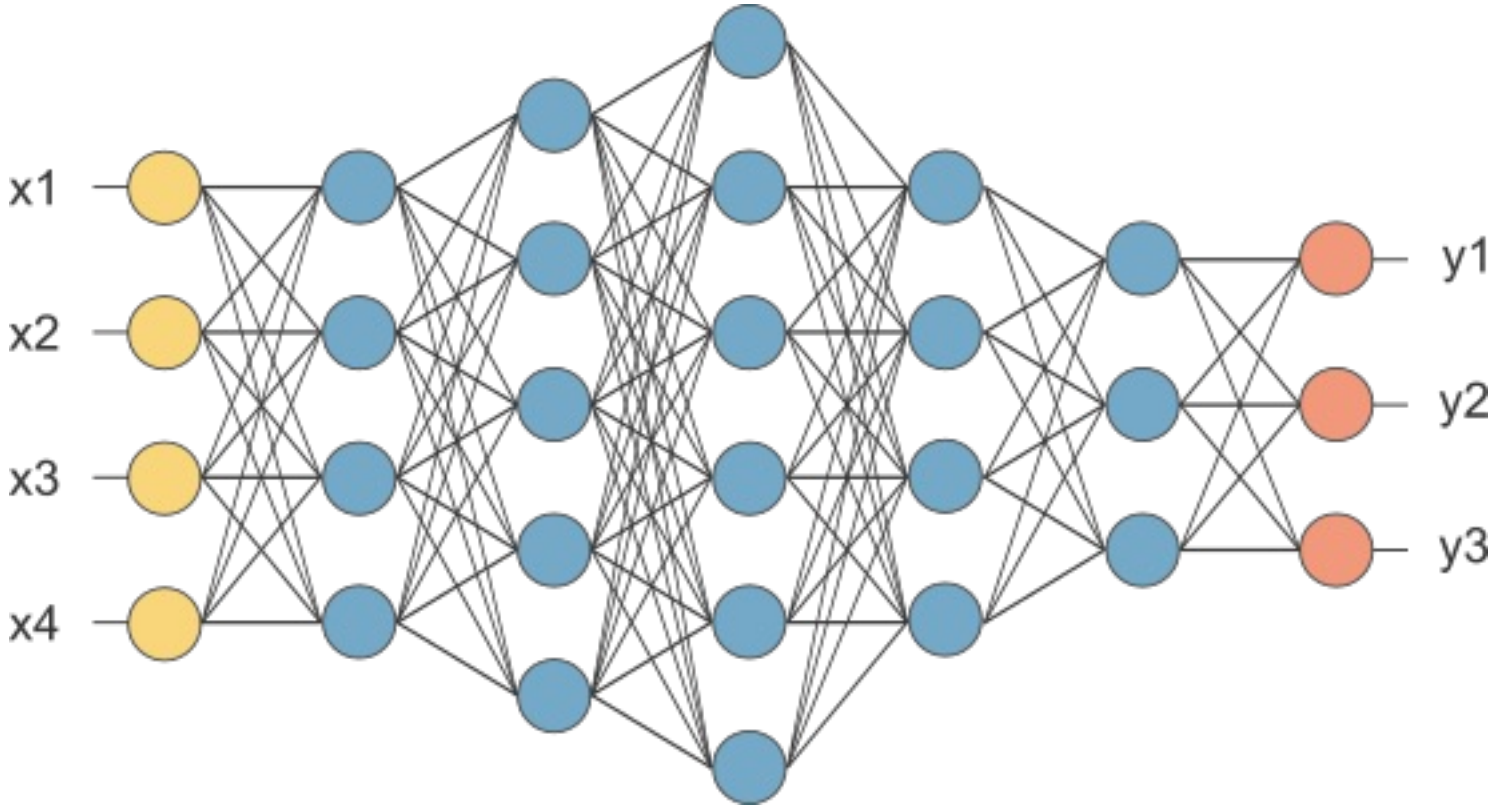
LECTURE TWO


# Attention and Transformers

Plan: We trace back history to see how attention and transformers have emerged

1. Basic models, related to transduction models and attention.
2. Encoder-Decoder model, using recurrent networks such as LSTM.
3. Transformer models are general models sufficient for almost all biotech applications (graph models may be treated to be special cases too).
4. For example, DeepMind AlphaFold2 uses depends on a transformer architecture to train an end-to-end model.
5. The transformer model also makes it easy for large scale biological data (pre)training.

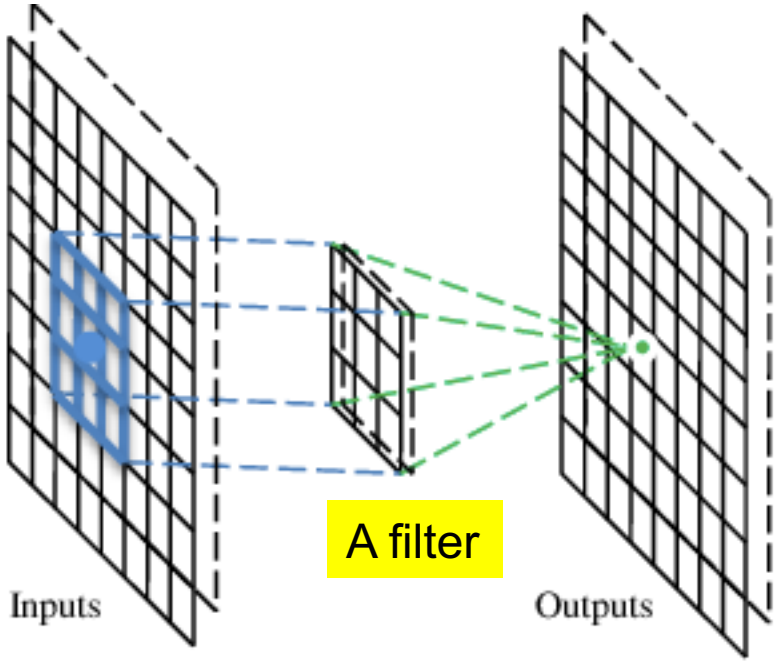
# 1. Fully connected network, feedforward network




To learn the weights on the edges

# 2. CNN

A CNN is a neural network with some convolutional layers (and some other layers). A convolutional layer has a number of filters that do convolutional operation.



# Convolutional layer

| | | | | | |
|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 |

Input

**These are the network parameters to be learned.**

| | | |
|---|---|---|
| 1 | -1 | -1 |
| -1 | 1 | -1 |
| -1 | -1 | 1 |

Filter 1

| | | |
|---|---|---|
| -1 | 1 | -1 |
| -1 | 1 | -1 |
| -1 | 1 | -1 |

Filter 2

⋮ ⋮

Each filter detects a small pattern (3 x 3).

# Convolution Operation

| | | |
|---|---|---|
| 1 | -1 | -1 |
| -1 | 1 | -1 |
| -1 | -1 | 1 |

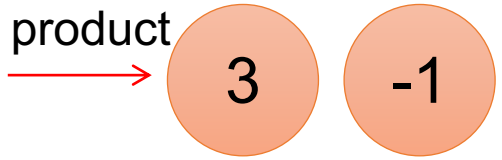
Filter 1

stride=1

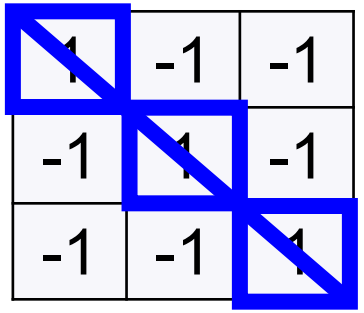
| | | | | | |
|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 |

Input

Dot product

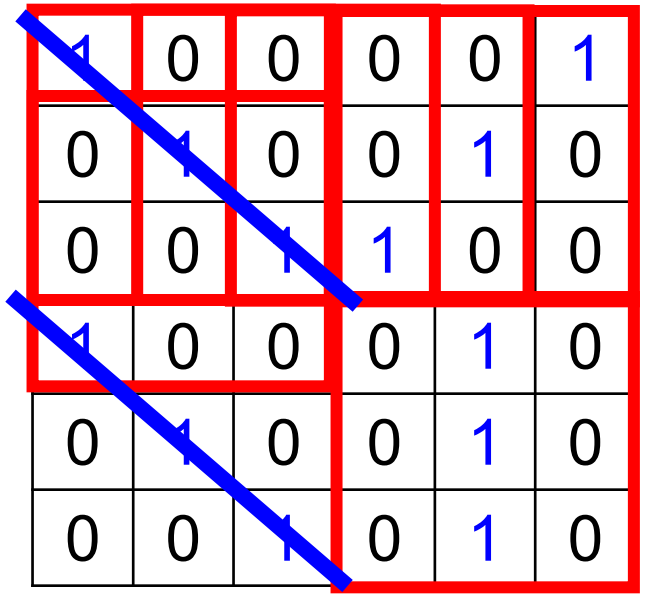


# Convolution

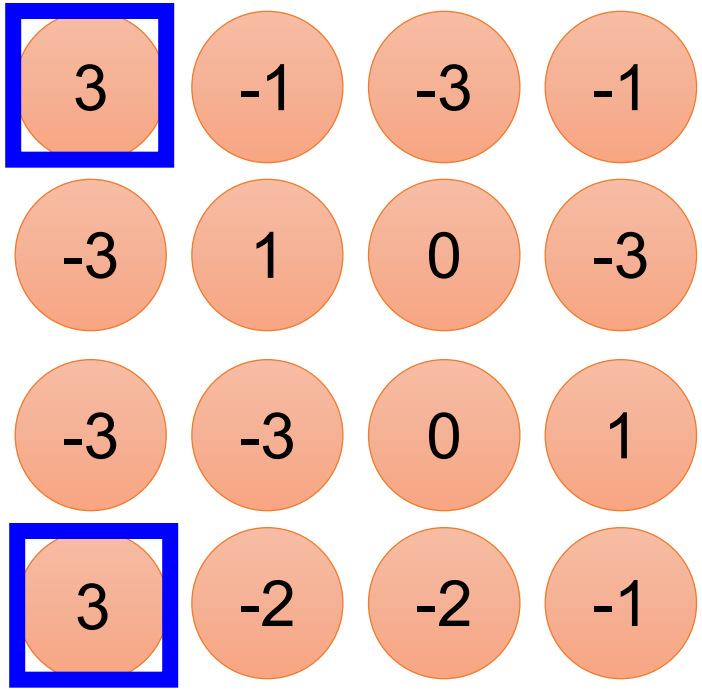



Filter 1

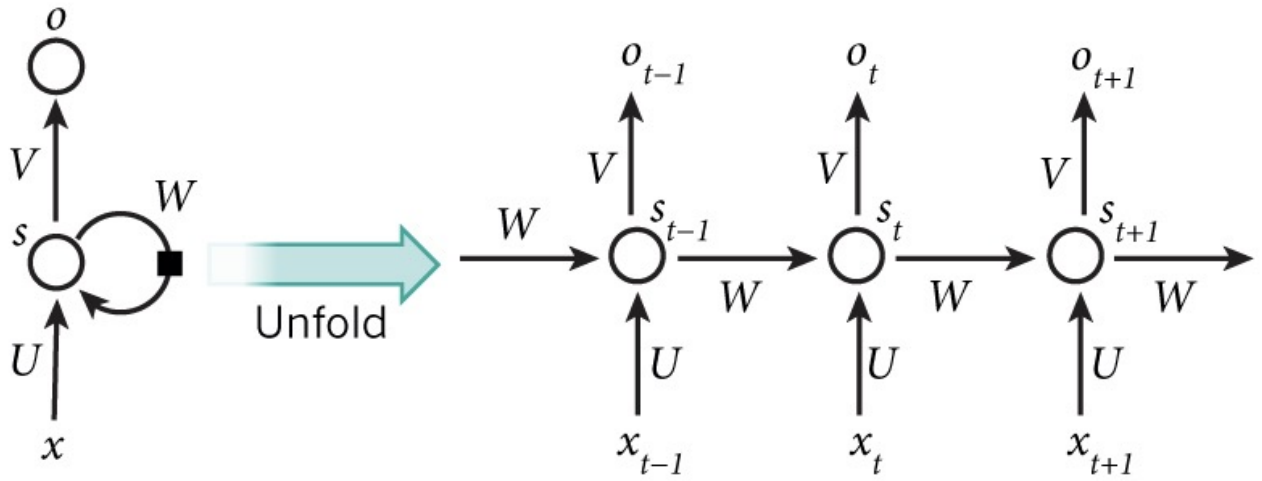
stride=1




Input

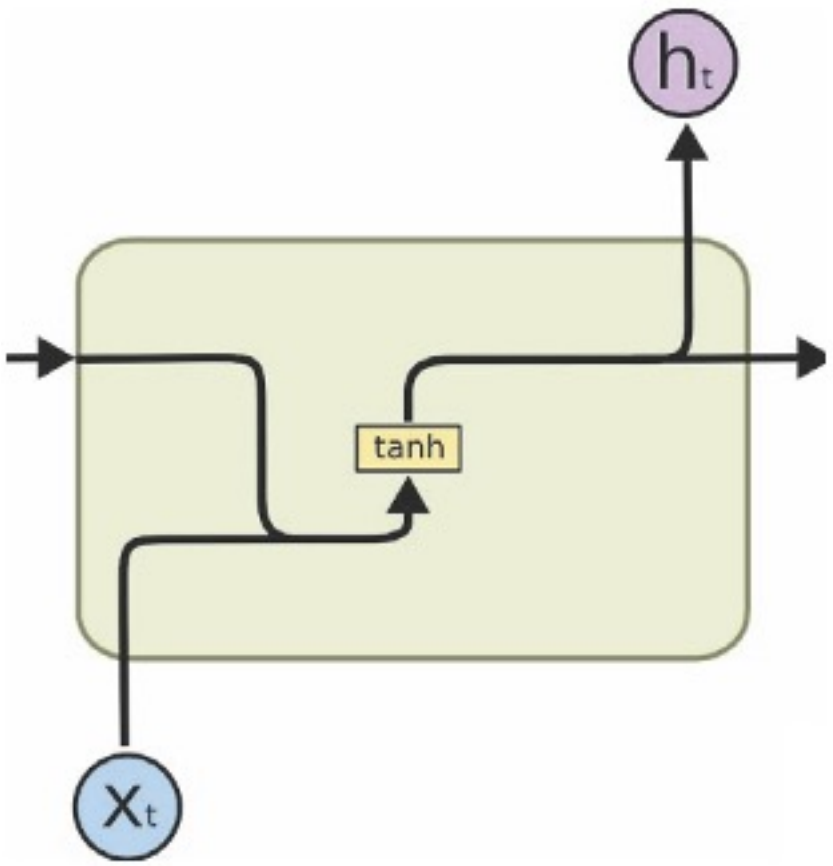


# 3. RNN

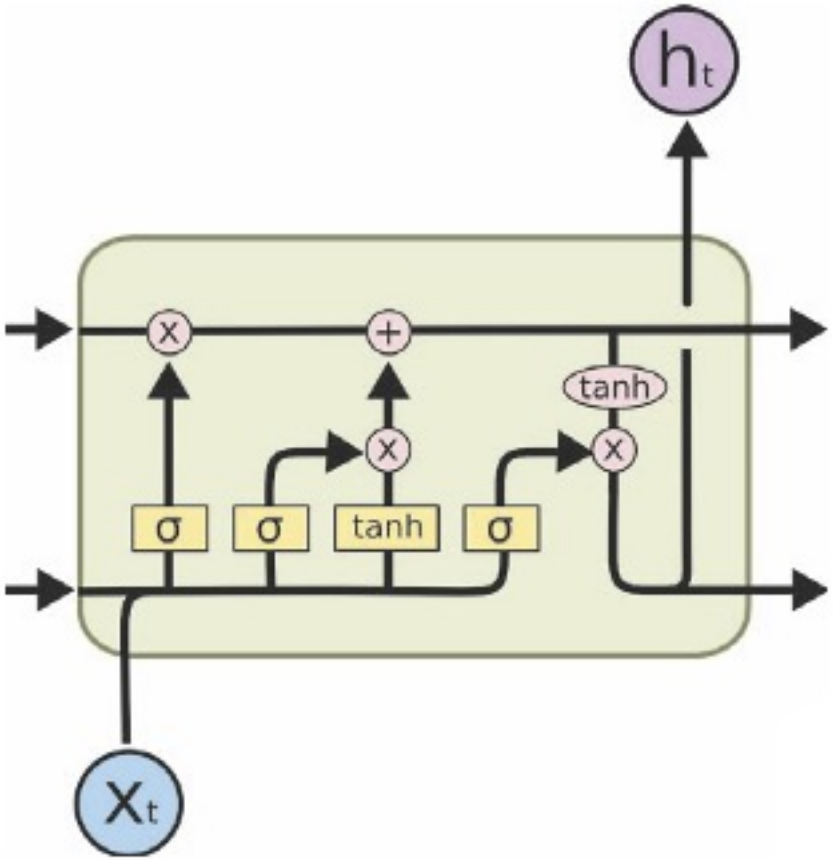



Parameters to be learned:
U, V, W

# Simple RNN vs LSTM

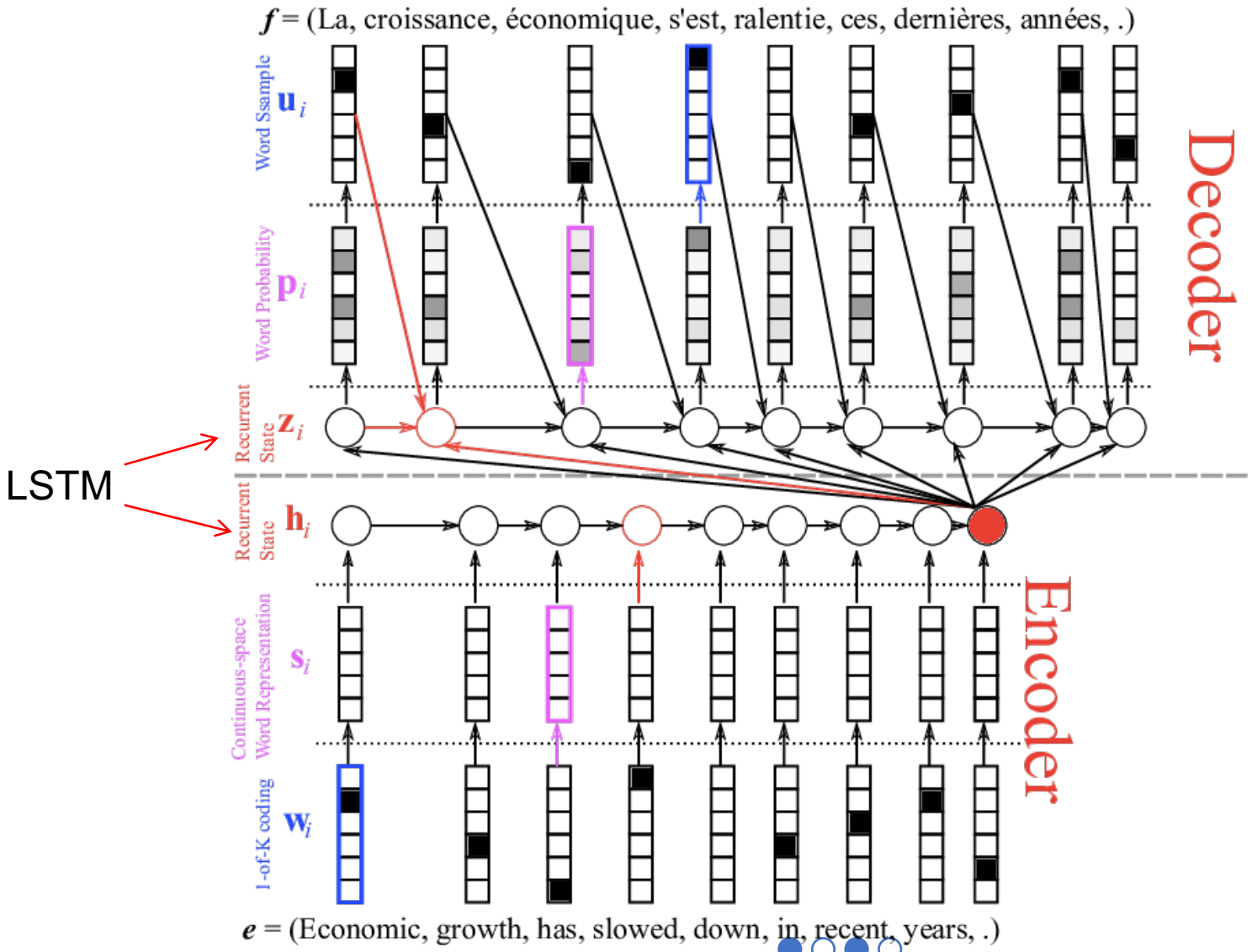



(a) RNN

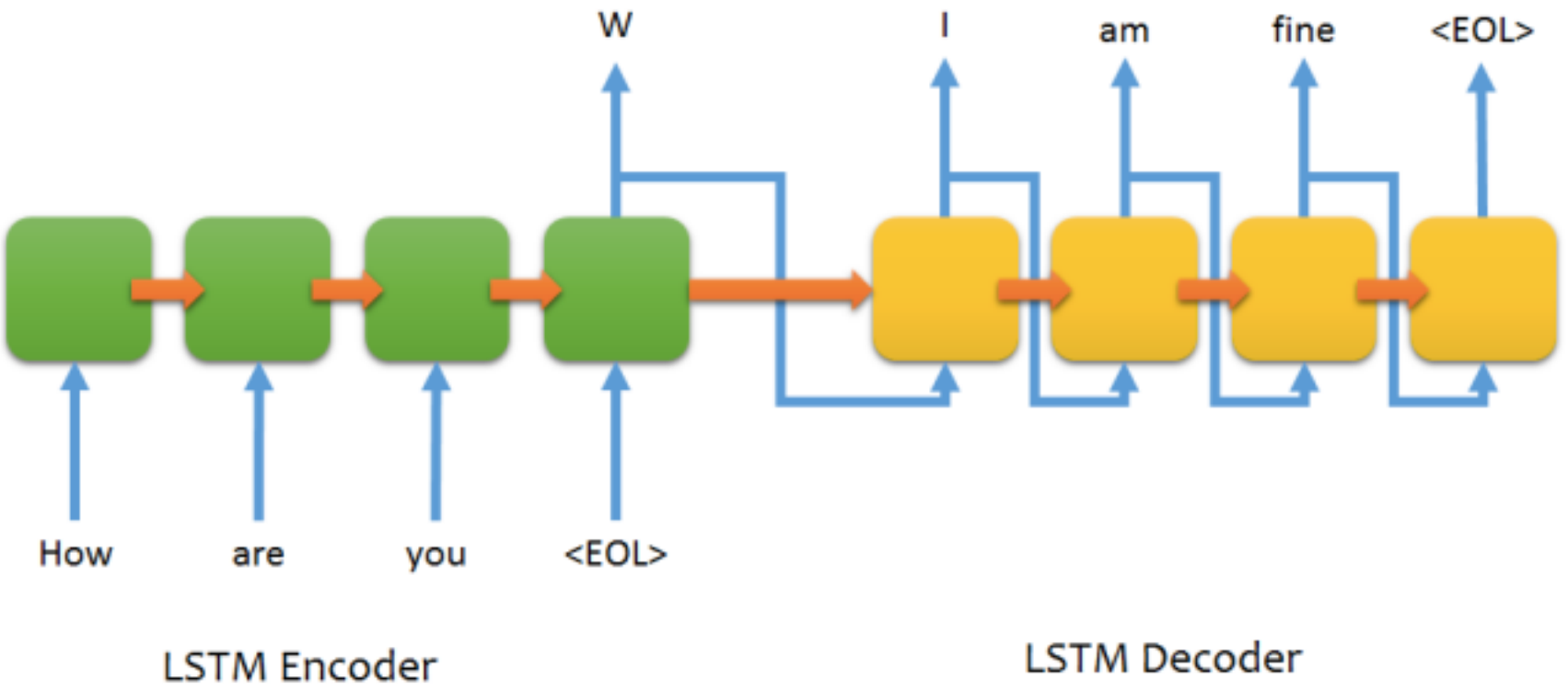



(b) LSTM

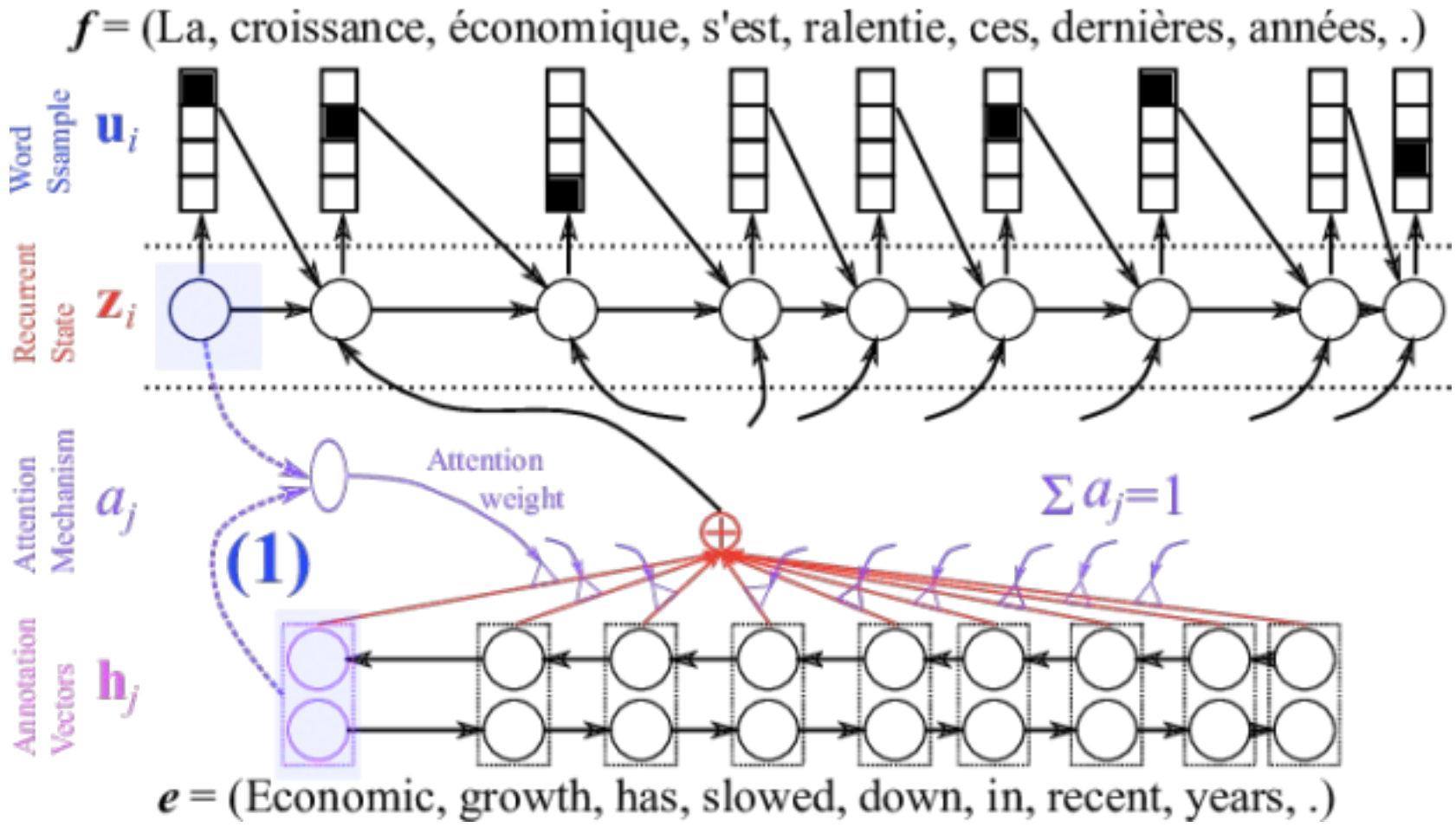
# Encoder-Decoder machine translation



# Encoder-Decoder LSTM structure for chatting



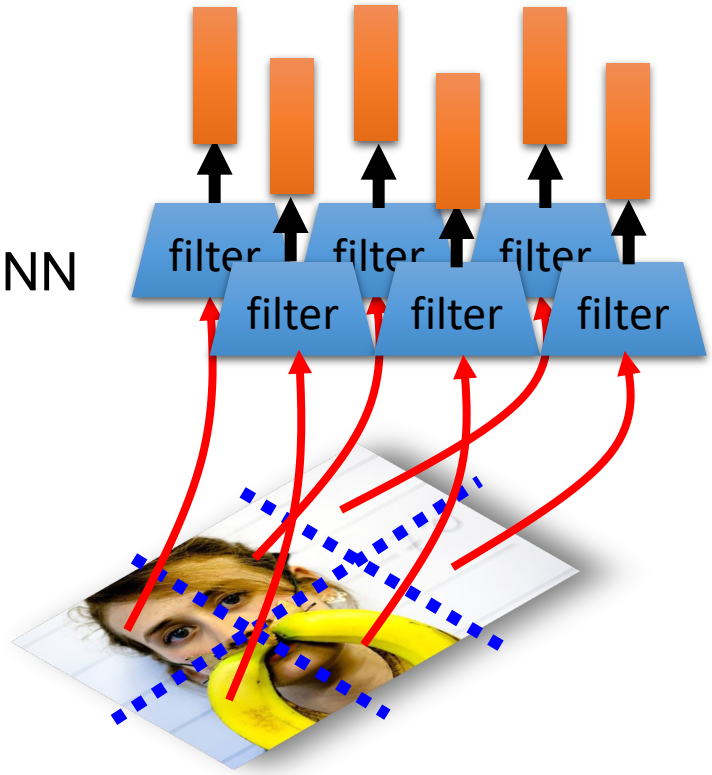
# Attention


f = (La, croissance, économique, s'est, ralentie, ces, dernières, années, .)
Word Ssample
$u_i$
Recurrent State
$z_i$
Attention Mechanism
$a_j$
Attention weight
$\sum a_j=1$
(1)
Annotation Vectors
$h_j$
e = (Economic, growth, has, slowed, down, in, recent, years, .)

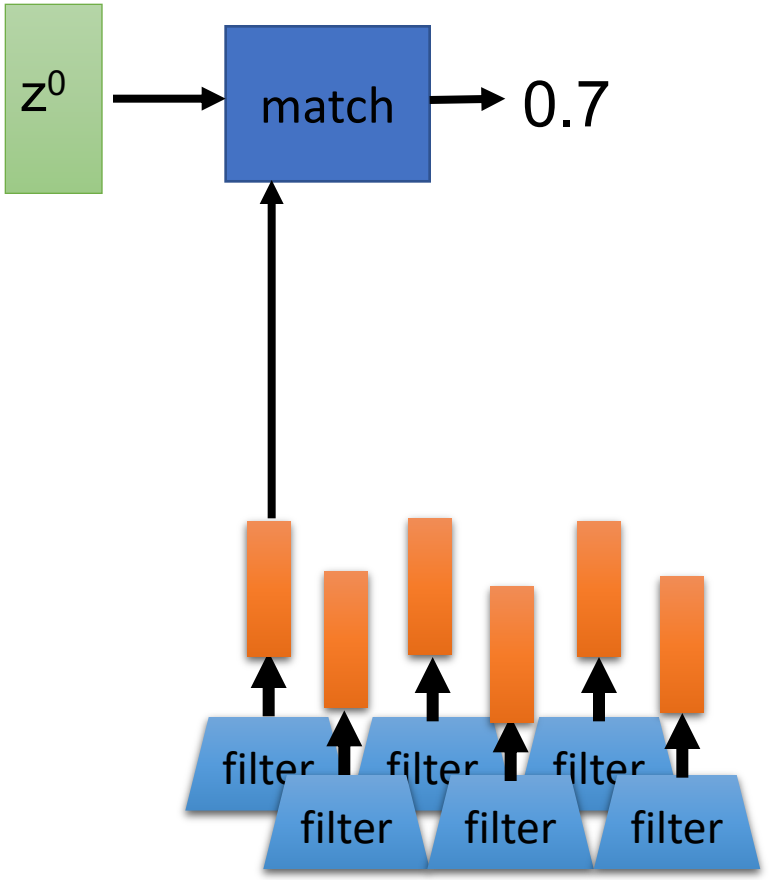
# Image caption generation using attention

A vector for each region

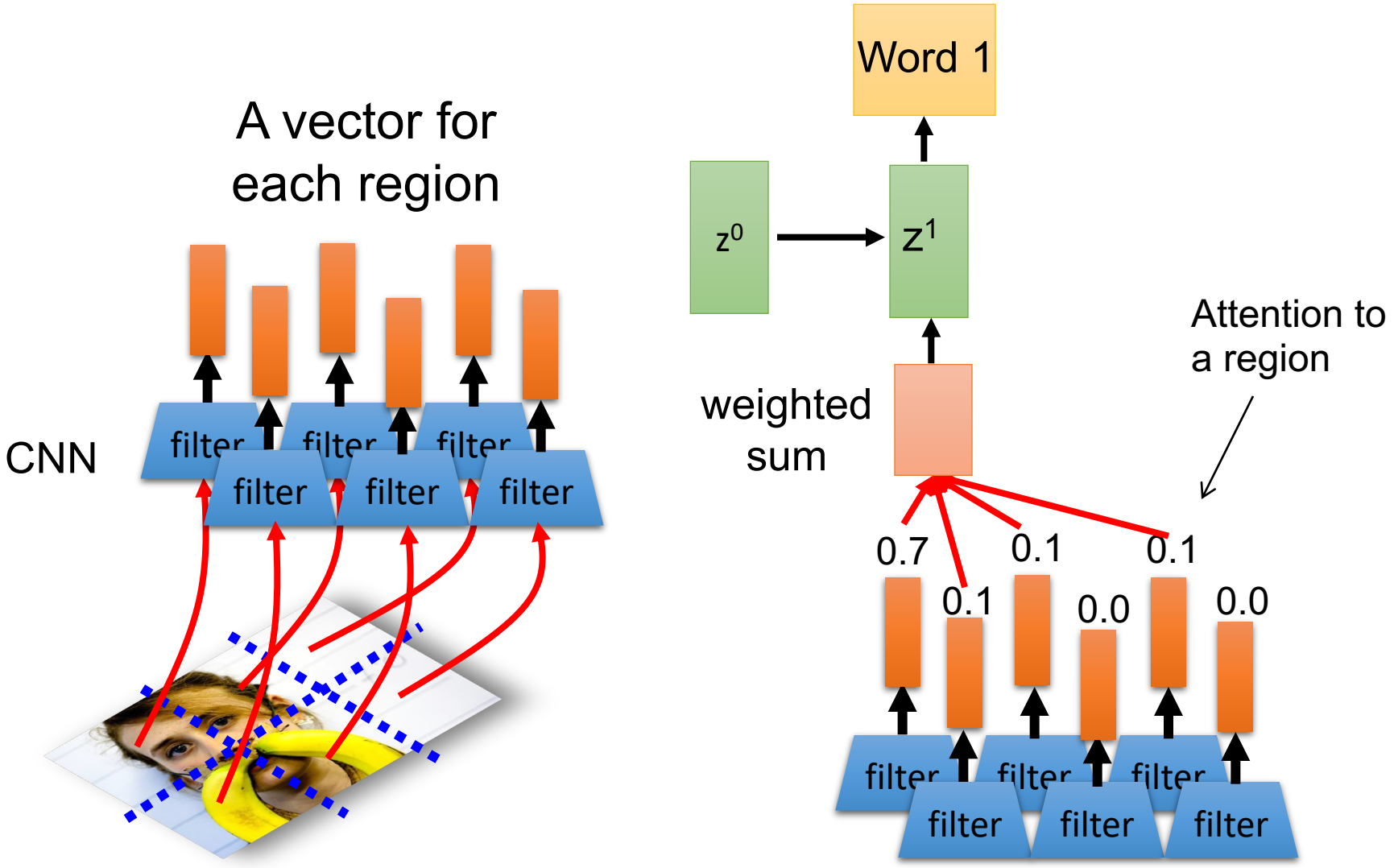
CNN



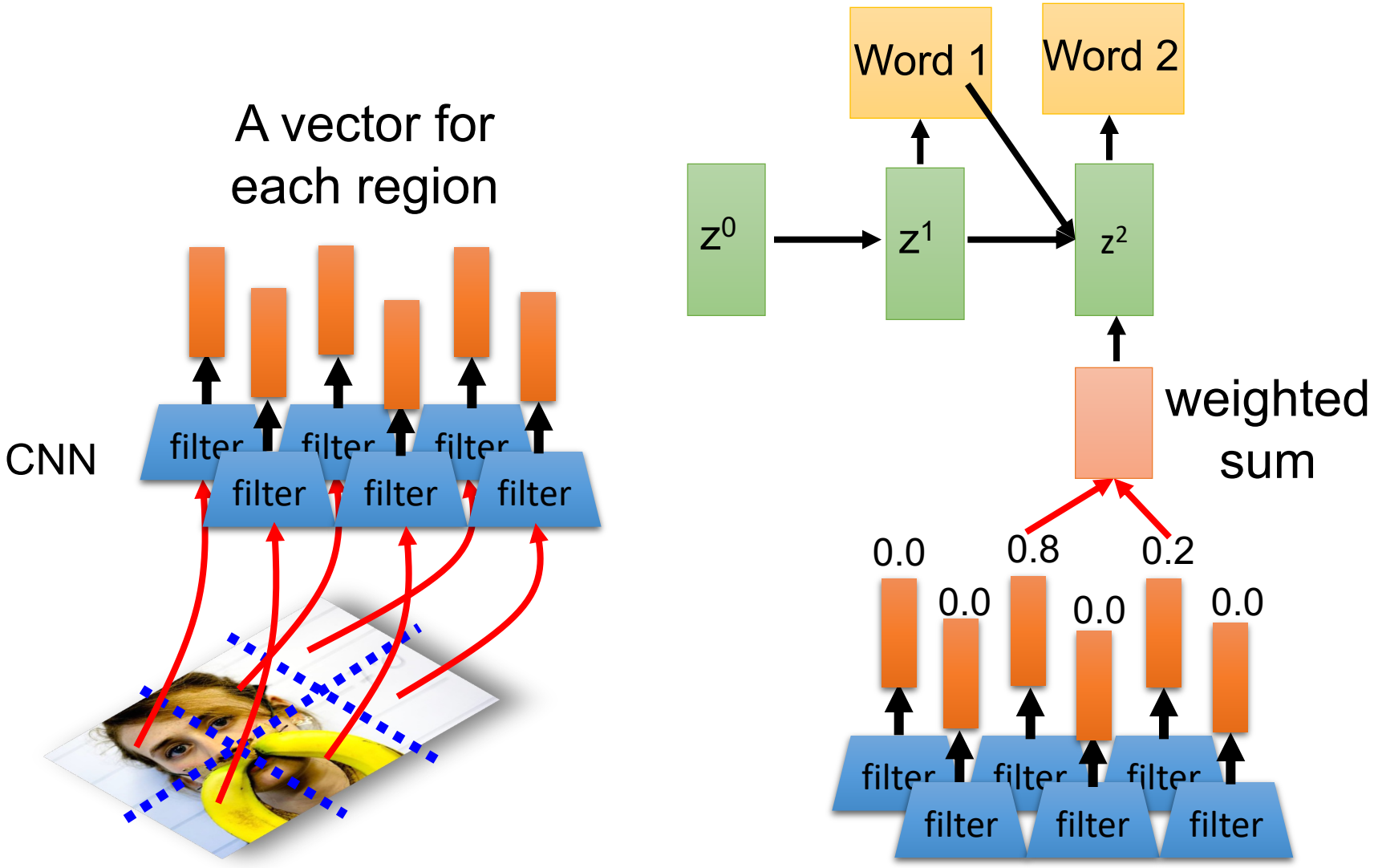

$z^0$ is initial parameter, it is also learned



# Image caption generation using attention



# Image caption generation using attention



# Image caption generation using attention

A woman is throwing a frisbee in a park.

A dog is standing on a hardwood floor.


A stop sign is on a road with a mountain in the background.

A little girl sitting on a bed with a teddy bear.

A group of people sitting on a boat in the water.

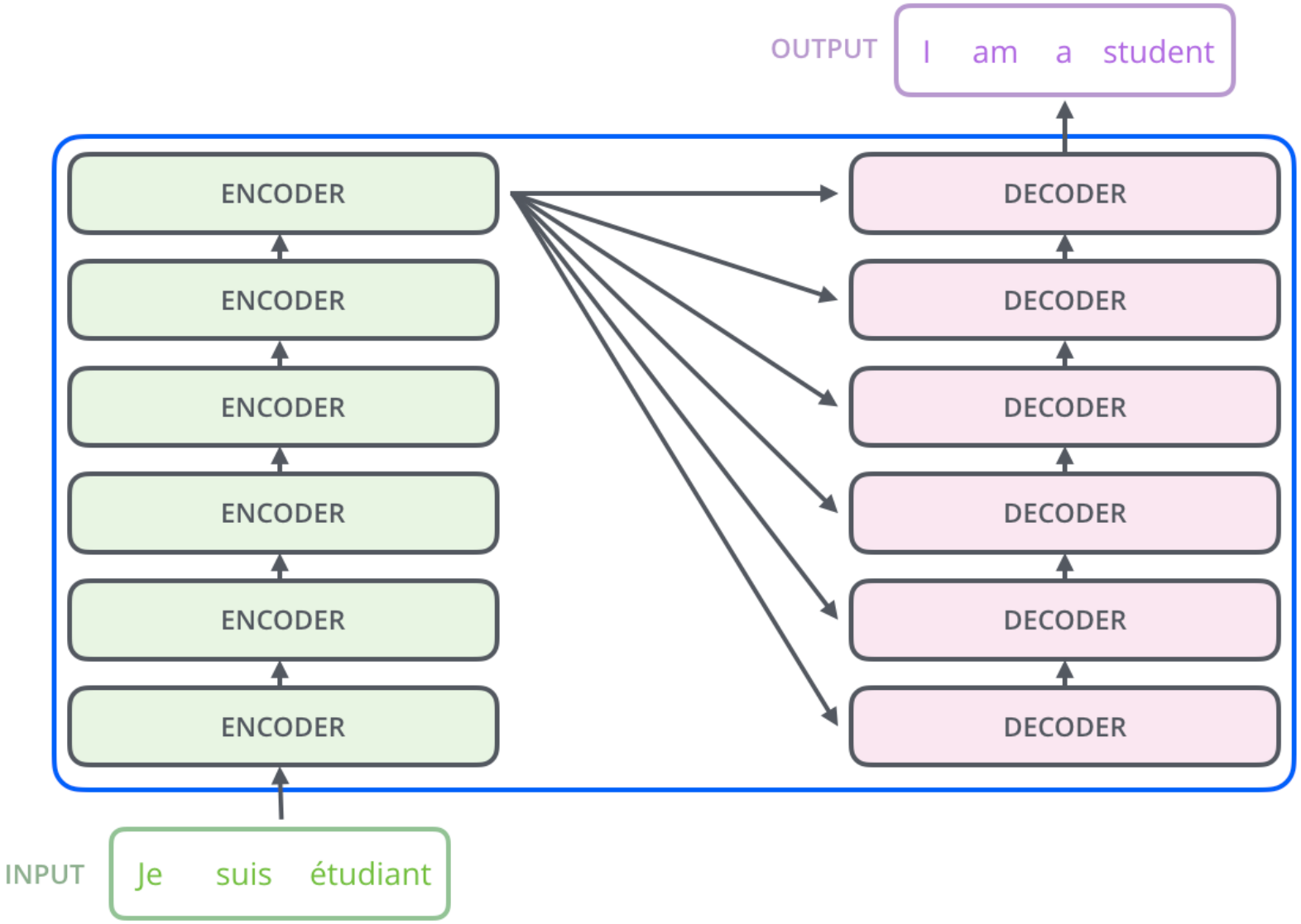
A giraffe standing in a forest with trees in the background.

Kelvin Xu, Jimmy Ba, Ryan Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhutdinov, Richard Zemel, Yoshua Bengio, “Show, Attend and Tell: Neural Image Caption Generation with Visual Attention”, ICML, 2015

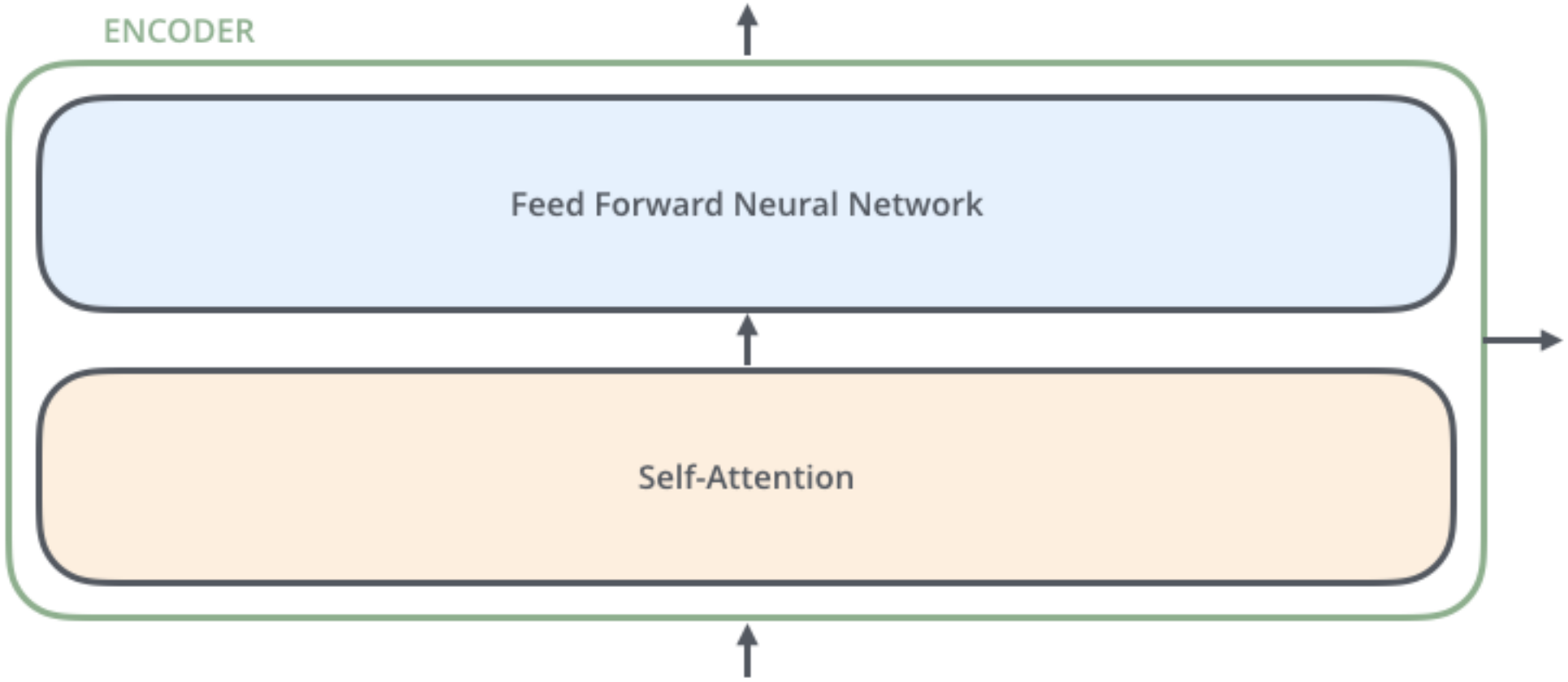
# More new ideas:

1. ULM-FiT, pre-training, transfer learning in NLP
2. Recurrent models require linear sequential computation, hard to parallelize. ELMo, bidirectional LSTM.
3. In order to reduce such sequential computation, several models based on CNN are introduced, such as ConvS2S and ByteNet. Dependency for ConvS2S needs linear depth, and ByteNet logarithmic.
4. The transformer is the first transduction model relying entirely on self-attention to compute the representations of its input and output without using RNN or CNN.

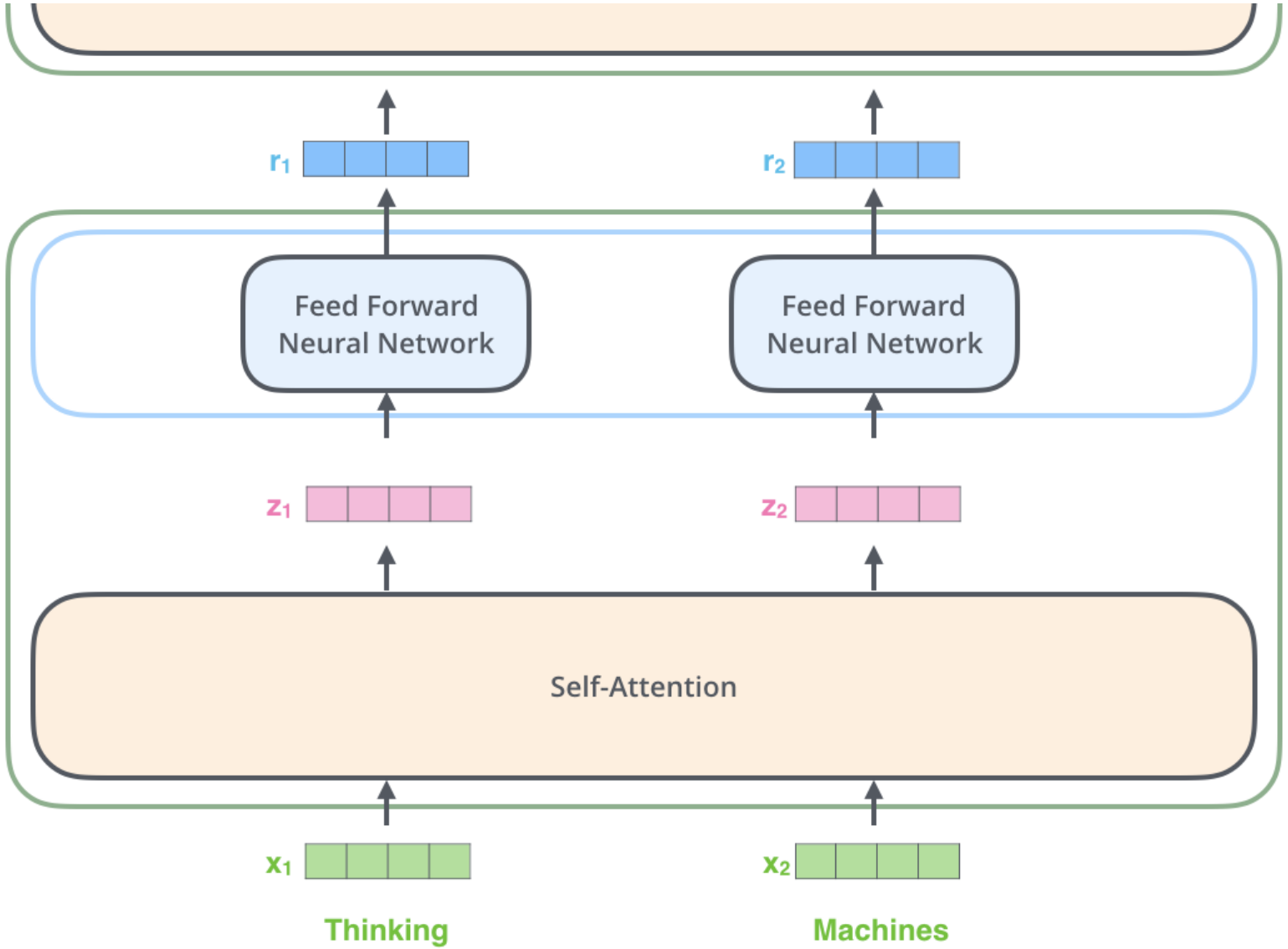
# Transformer



# An Encoder Block: same structure, different parameters

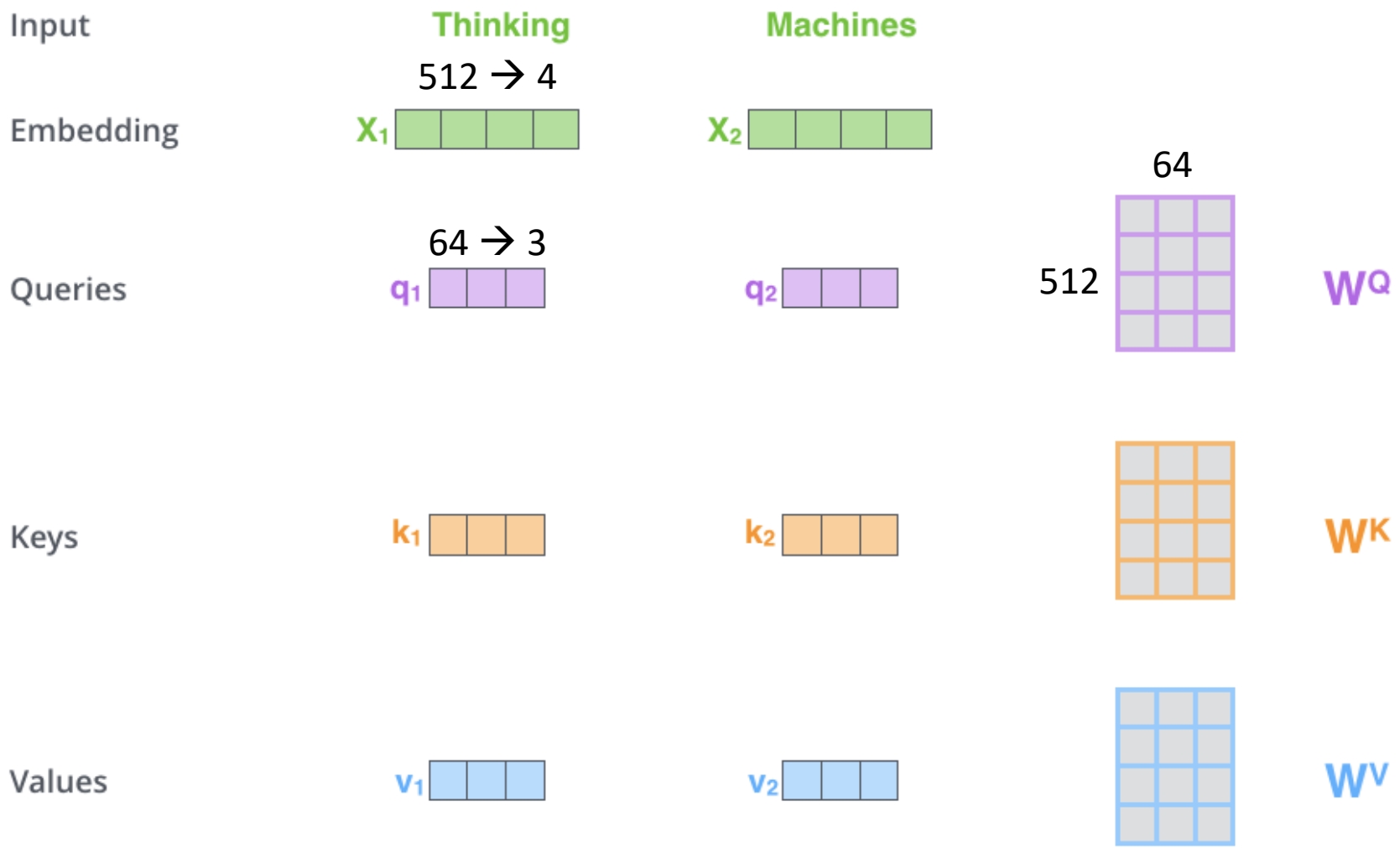


# Encoder




Note: The ffnn is independent for each word.
Hence can be parallelized.

# Self Attention

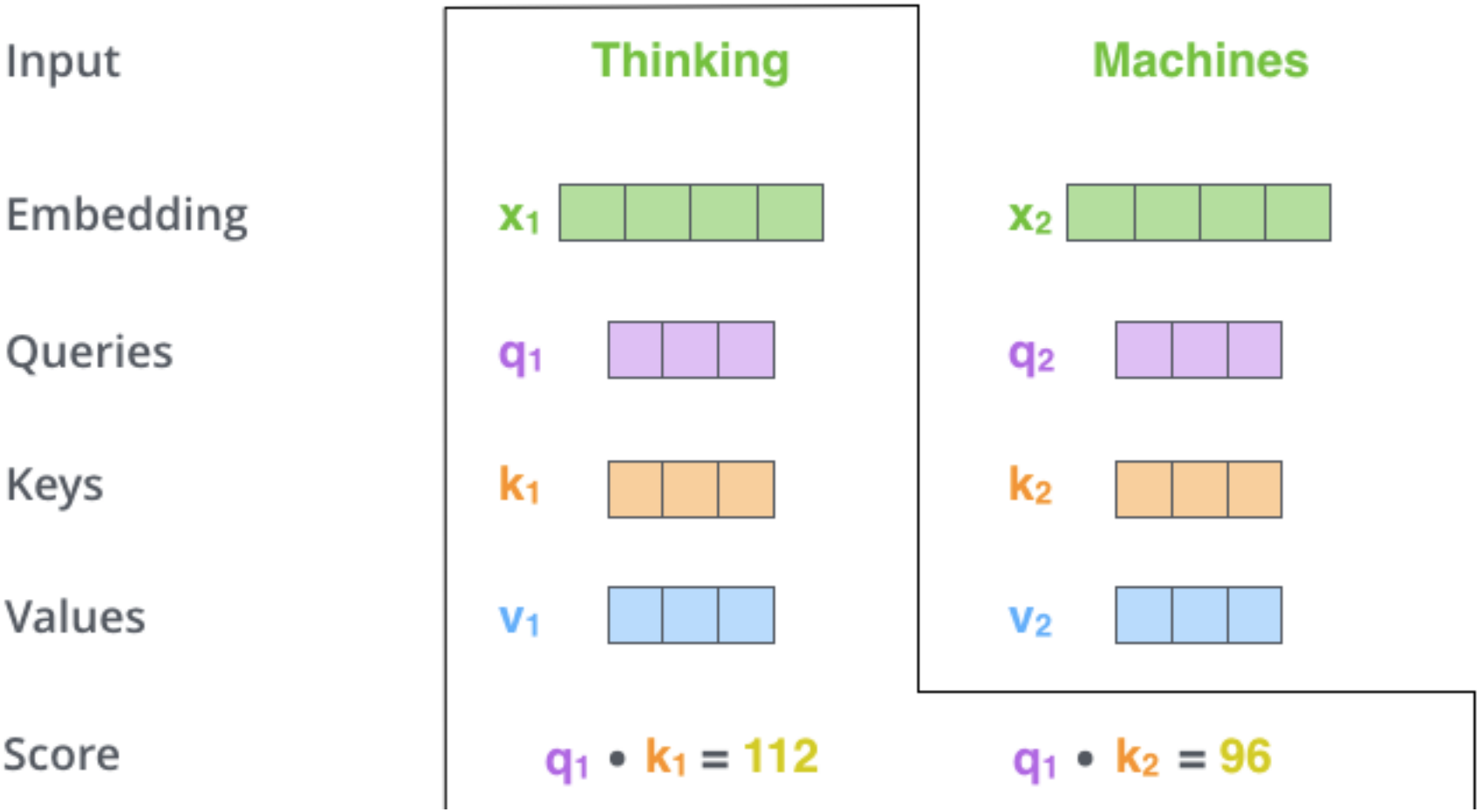



First we create three vectors by multiplying input embedding (1x512)
$x_i$ with three matrices (64x512):

$q_i = x_i W^Q$
$K_i = x_i W^K$
$V_i = x_i W^V$

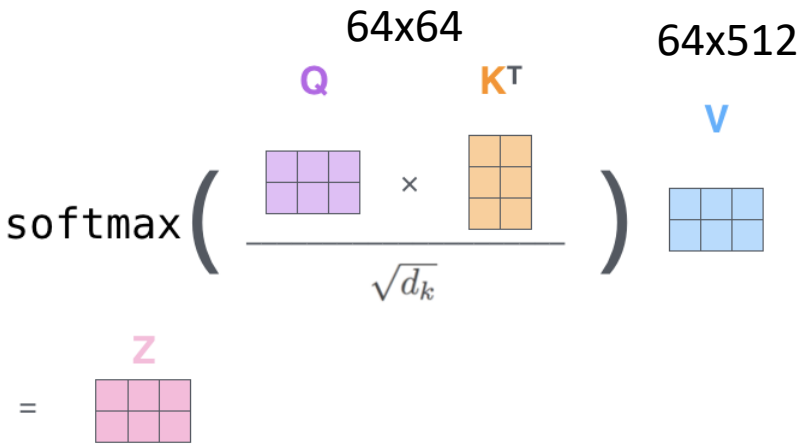
# Self Attention

Now we need to calculate a score to determine how much focus to place on other Parts of the input.

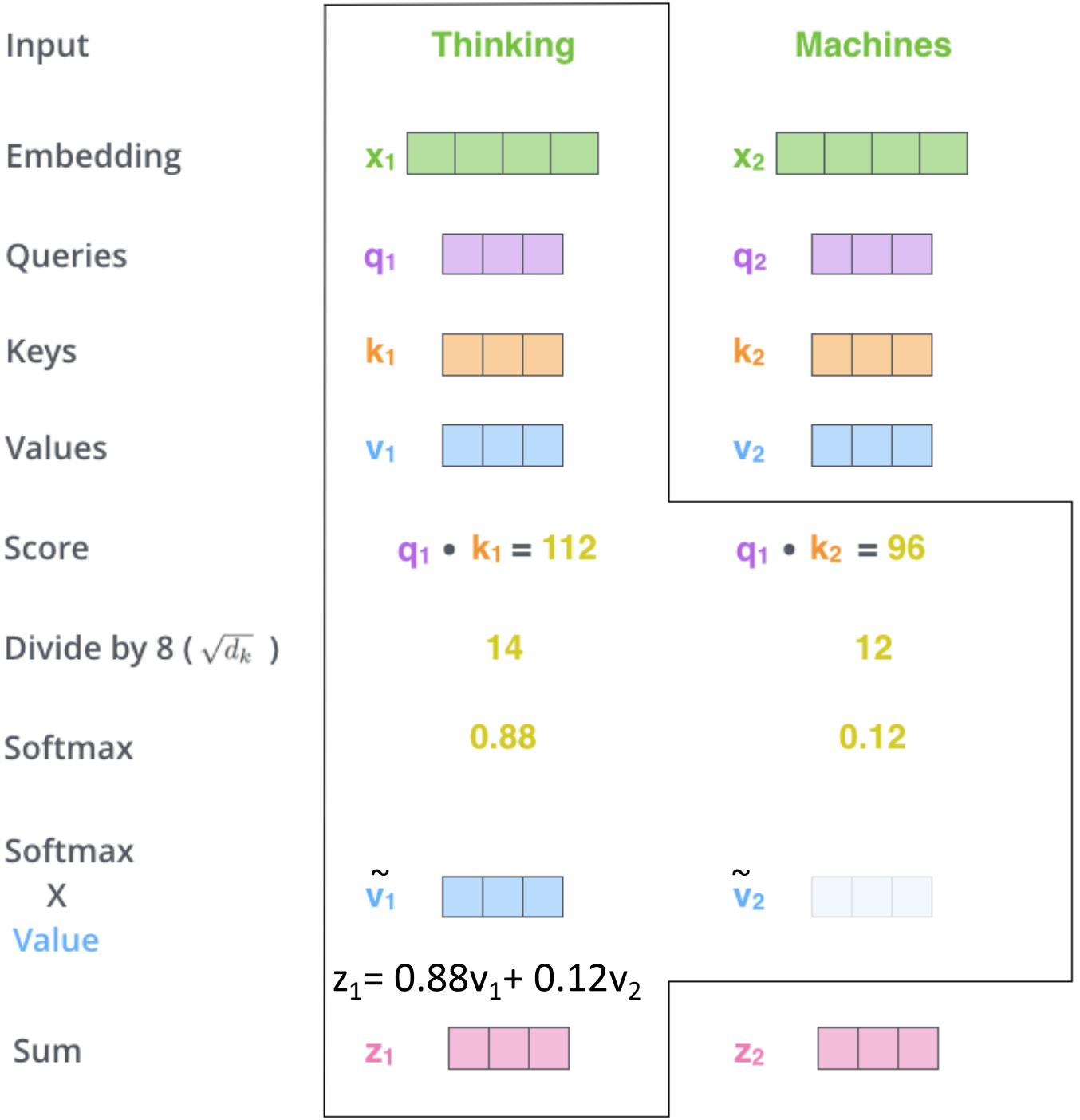


# Self Attention

Formula

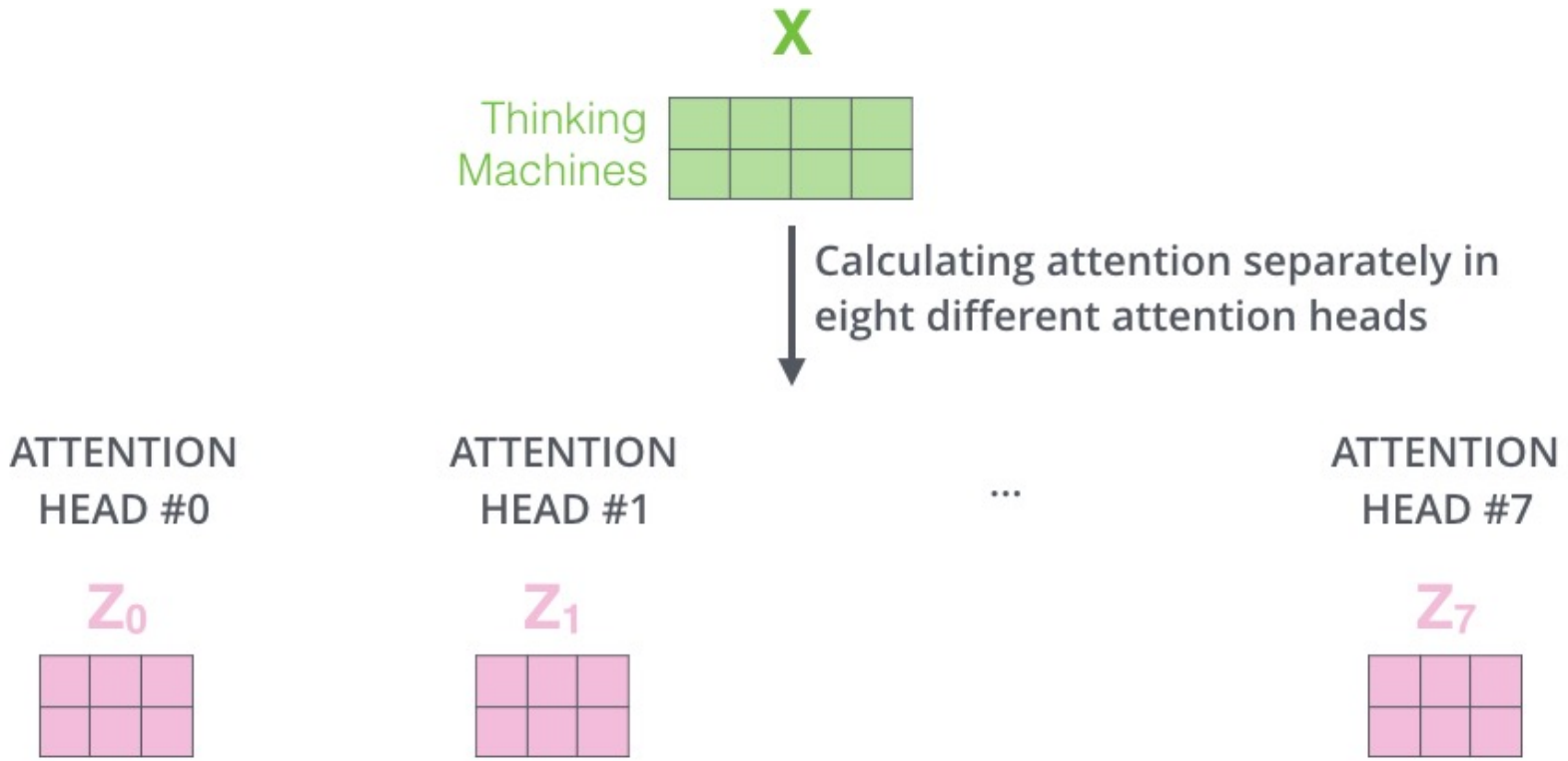



$d_k$=64 is dimension of key vector



# Multiple heads

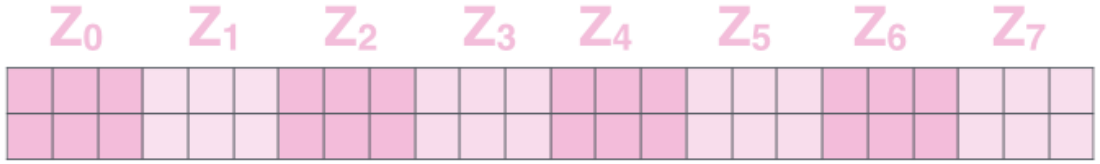
1. It expands the model's ability to focus on different positions.
2. It gives the attention layer multiple "representation subspaces"



# Attention and Transformers

The output is expecting only a 2x4 matrix, hence,

1) Concatenate all the attention heads




2) Multiply with a weight matrix $W^O$ that was trained jointly with the model




3) The result would be the Z matrix that captures information from all the attention heads. We can send this forward to the FFNN

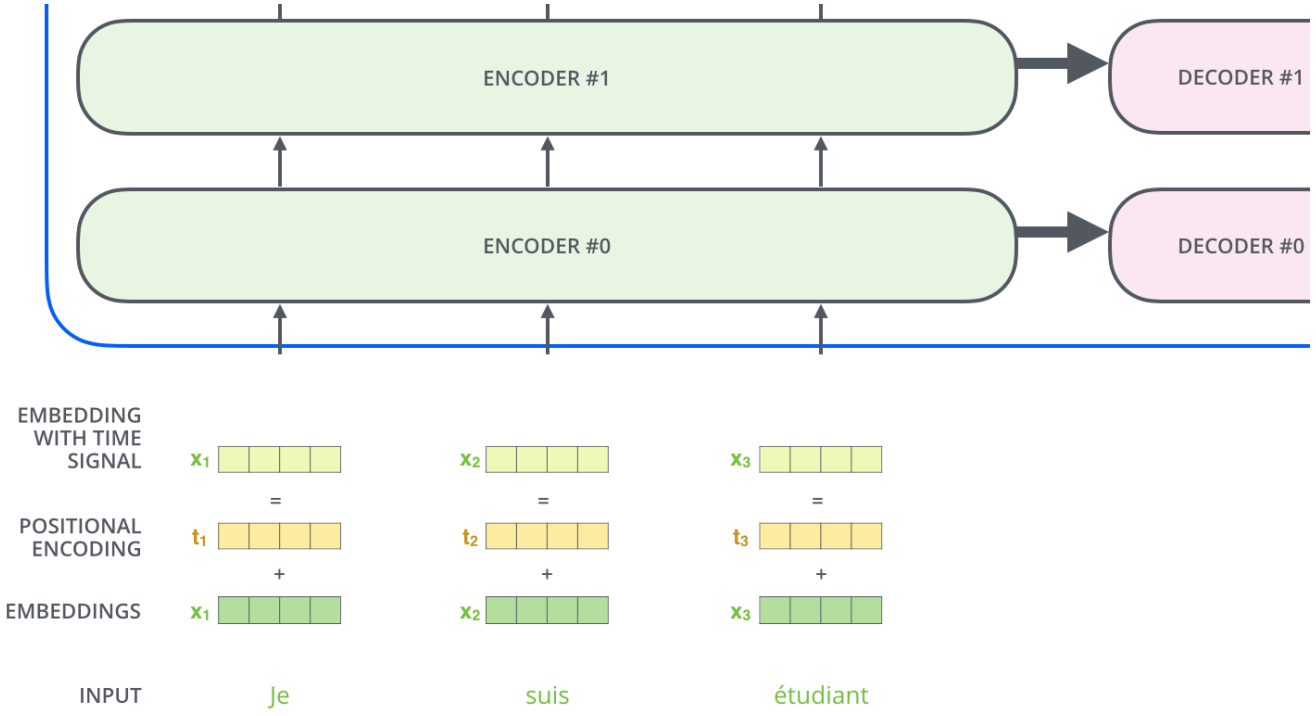



If you want some more intuition on attention: watch https://www.youtube.com/watch?v=-9vVhYEXeyQ

# Representing the input order (positional encoding)

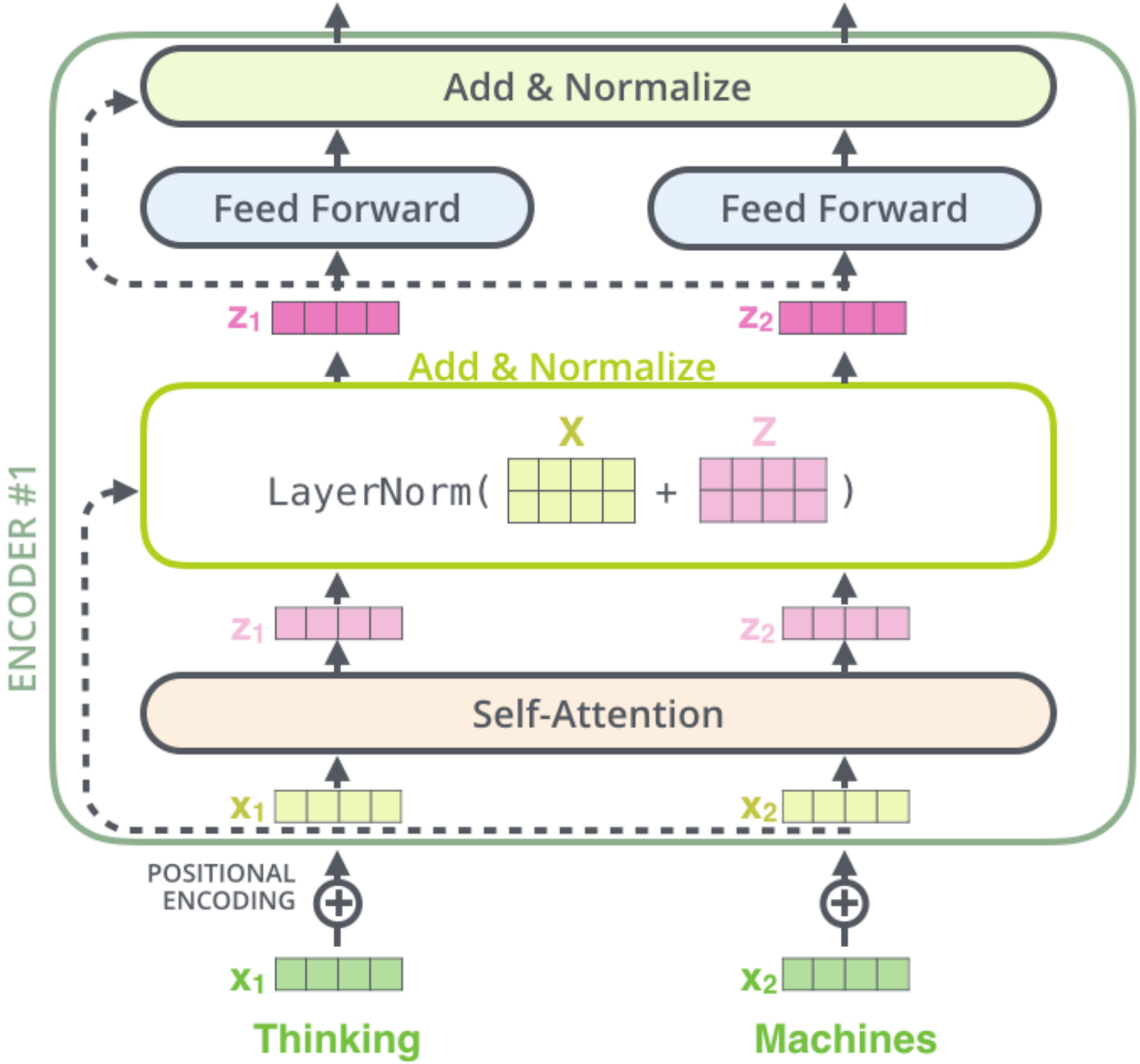
The transformer adds a vector to each input embedding. These vectors follow a specific pattern that the model learns, which helps it determine the position of each word, or the distance between different words in the sequence. The intuition here is that adding these values to the embedings provides meaningful distances between the embedding vectors once they're projected into Q/K/V vectors and during dot-product attention.

More on positional encoding:
https://kazemnejad.com/blog/transformer_architecture_positional_encoding/



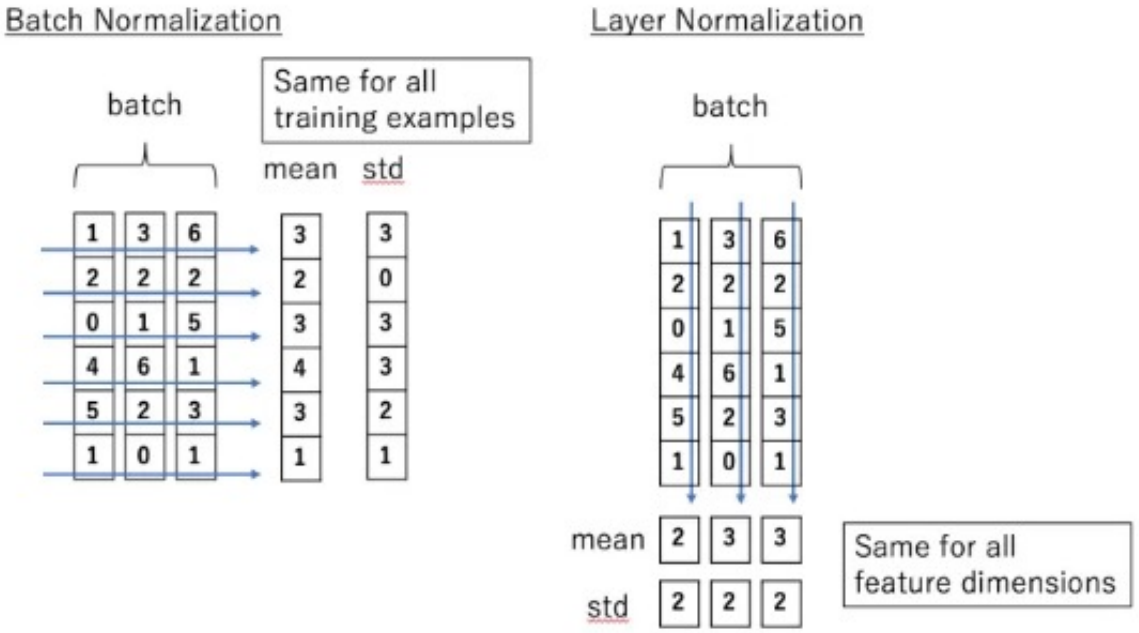
# Add and Normalize

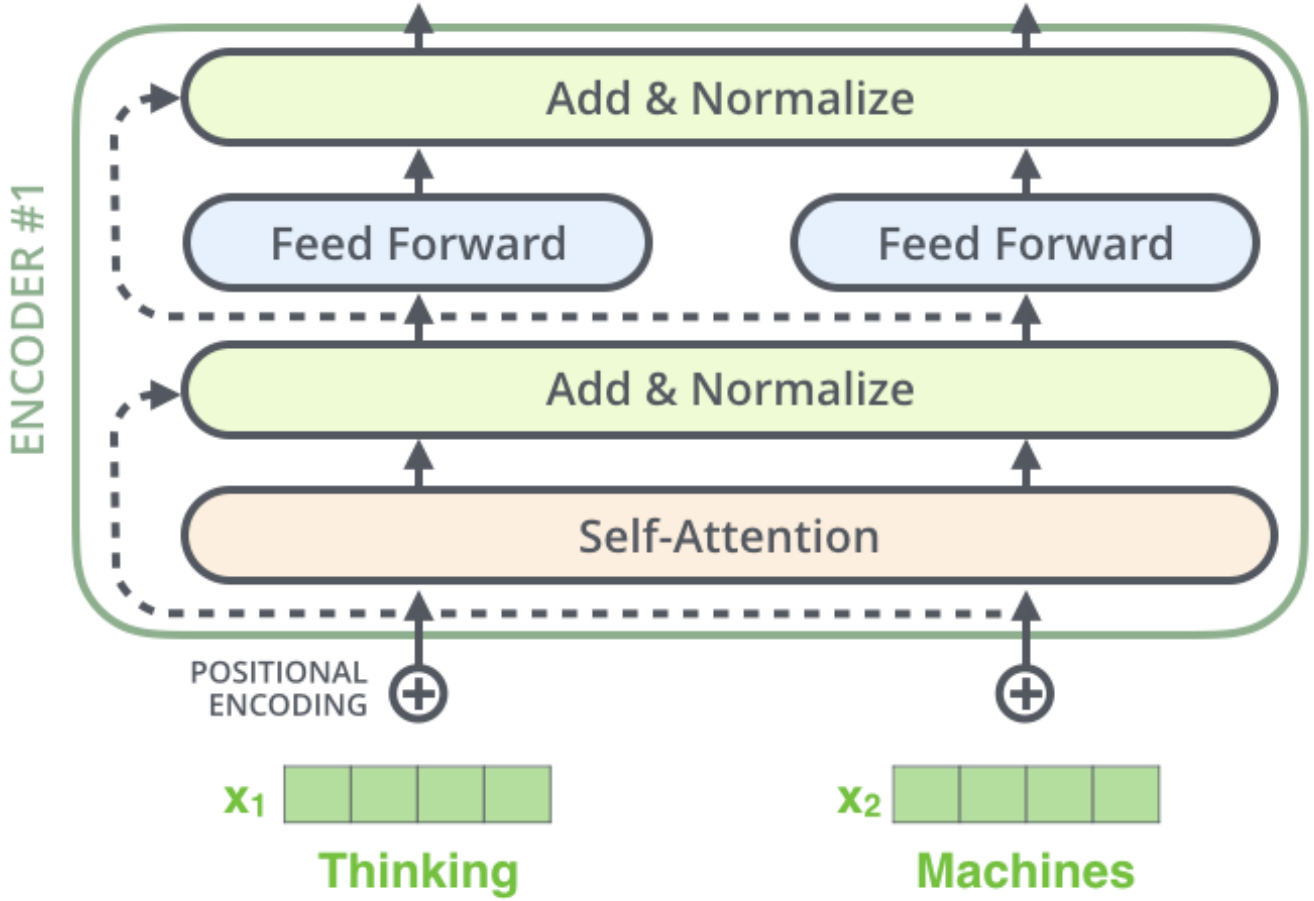
In order to regulate the computation, this is a normalization layer so that each feature (column) have the same average and deviation.



# Layer Normalization (Hinton)

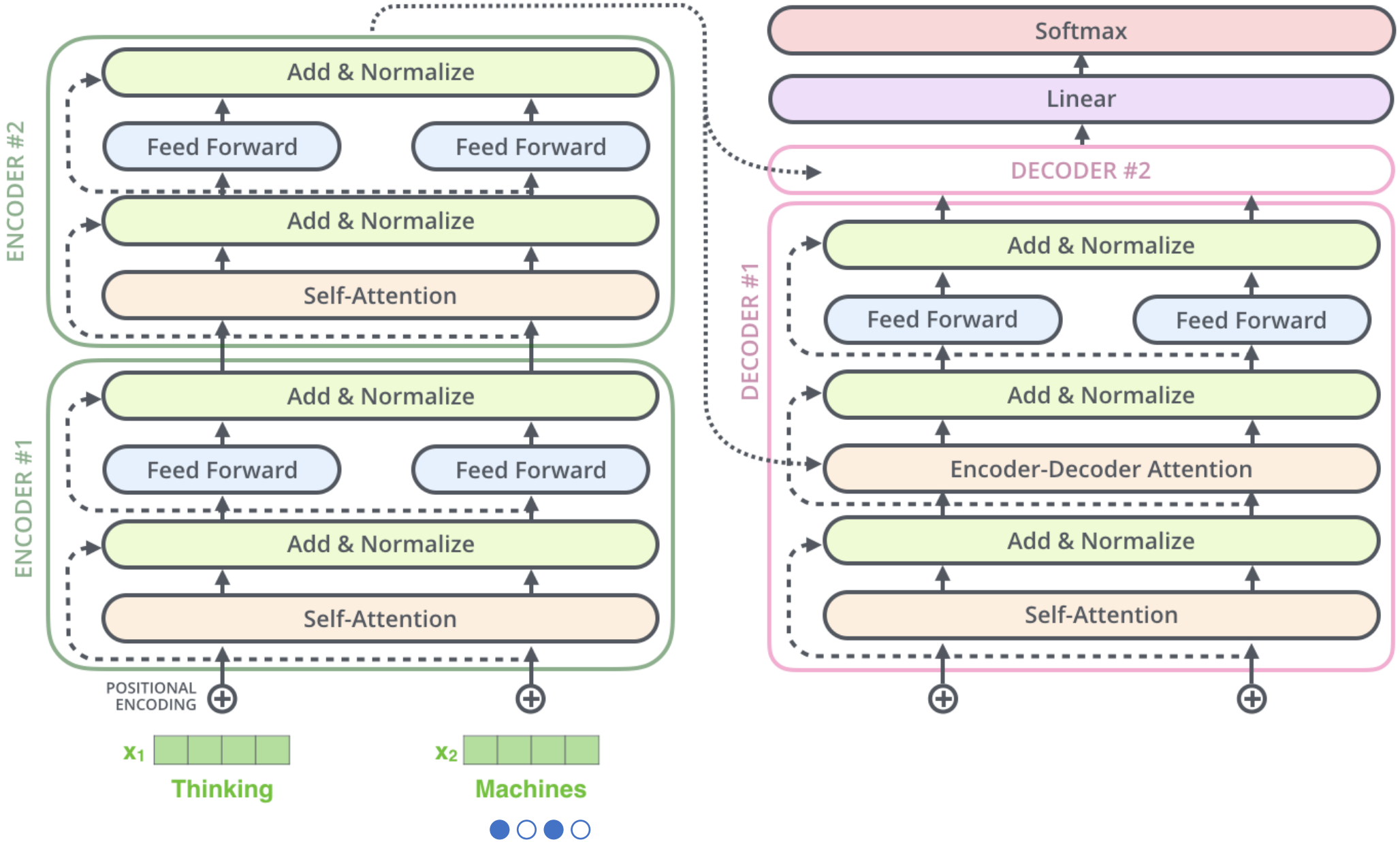
Layer normalization normalizes the inputs across the features.



# The complete transformer

The encoder-decoder attention is just like self attention, except it uses K, V from the top of encoder output, and its own Q



Note: In decoder, the input is “incomplete” when calculating self-attention.

The solution is to set future unknown values with “-inf”.

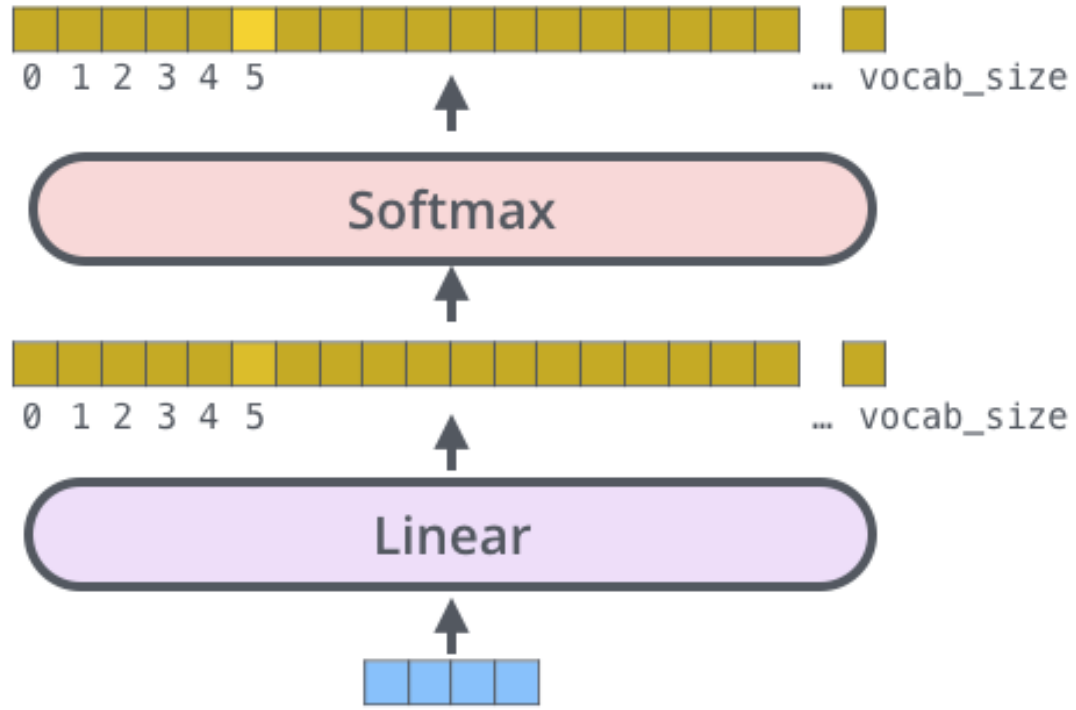
Decoder's Output Linear Layer

Which word in our vocabulary is associated with this index?

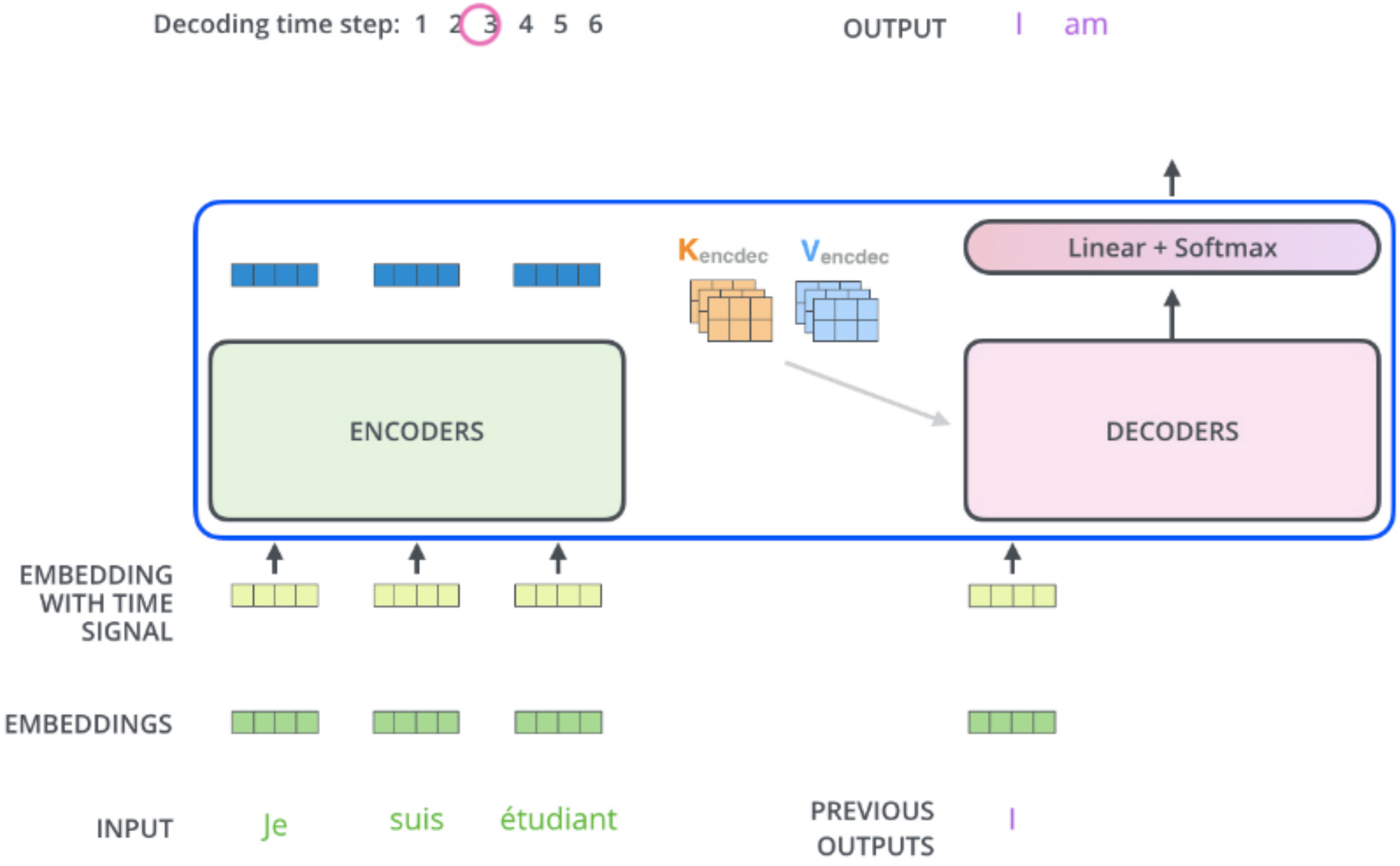
am

Get the index of the cell with the highest value (**argmax**)

5



# How it works




But what about Self-attention?

# Training and the Loss Function

Untrained Model Output

| 0.2 | 0.2 | 0.1 | 0.2 | 0.2 | 0.1 |
|---|---|---|---|---|---|

Correct and desired output

| 0.0 | 0.0 | 0.0 | 1.0 | 0.0 | 0.0 |
|---|---|---|---|---|---|
| a | am | I | thanks | student | <eos> |


We can use cross Entropy.

We can also optimize two words at a time: using BEAM search: keep a few alternatives for the first word.

# Cross Entropy and KL (Kullback-Leibler) divergence

- Entropy: $E(P) = -\Sigma_i P(i) \log P(i)$ - expected prefix-free code length (also optimal)
- Cross Entropy: $C(P) = -\Sigma_i P(i) \log Q(i)$ – expected coding length using optimal code for Q
- KL divergence:

  $D_{KL}(P \,||\, Q) = \Sigma_i P(i) \log[P(i)/Q(i)] = \Sigma_i P(i)[\log P(i) - \log Q(i)]$, extra bits to code using Q rather than P

- JSD$(P||Q) = \frac{1}{2} D_{KL}(P||M) + \frac{1}{2} D_{KL}(Q||M)$, $M = \frac{1}{2}(P+Q)$, symmetric KL

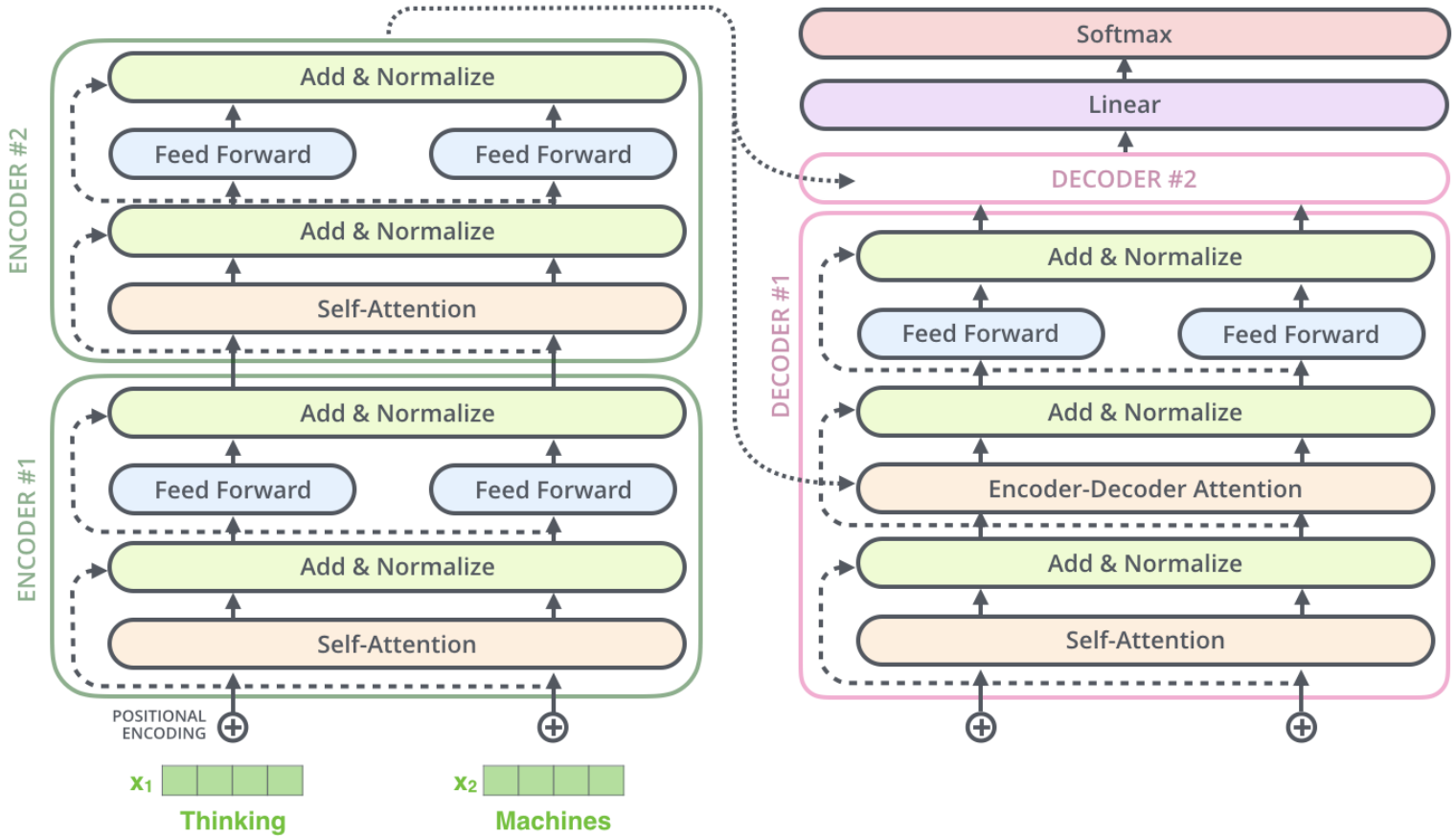
* JSD = Jensen-Shannon Divergency

# Transformer Results

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

| Model | BLEU | | Training Cost (FLOPs) | |
|---|---|---|---|---|
| | EN-DE | EN-FR | EN-DE | EN-FR |
| ByteNet [18] | 23.75 | | | |
| Deep-Att + PosUnk [39] | | 39.2 | | $1.0 \cdot 10^{20}$ |
| GNMT + RL [38] | 24.6 | 39.92 | $2.3 \cdot 10^{19}$ | $1.4 \cdot 10^{20}$ |
| ConvS2S [9] | 25.16 | 40.46 | $9.6 \cdot 10^{18}$ | $1.5 \cdot 10^{20}$ |
| MoE [32] | 26.03 | 40.56 | $2.0 \cdot 10^{19}$ | $1.2 \cdot 10^{20}$ |
| Deep-Att + PosUnk Ensemble [39] | | 40.4 | | $8.0 \cdot 10^{20}$ |
| GNMT + RL Ensemble [38] | 26.30 | 41.16 | $1.8 \cdot 10^{20}$ | $1.1 \cdot 10^{21}$ |
| ConvS2S Ensemble [9] | 26.36 | **41.29** | $7.7 \cdot 10^{19}$ | $1.2 \cdot 10^{21}$ |
| Transformer (base model) | 27.3 | 38.1 | **$3.3 \cdot 10^{18}$** | |
| Transformer (big) | **28.4** | **41.8** | $2.3 \cdot 10^{19}$ | |

# Next Lecture: Pretraining




BERT

GPT

# Literature & Resources for Transformers

Vaswani et al. Attention is all you need. 2017.

Resources:

https://nlp.seas.harvard.edu/2018/04/03/attention.html (Excellent explanation of transformer model with codes.)

Jay Alammar, The illustrated transformer (from which I borrowed many pictures):

http://jalammar.github.io/illustrated-transformer/